Comparing integers

Two rules, both read off the coordinate line — and one trap with negatives.

LESSON 23–24 | 2 × 40 MIN | MATEMATIKA 7, §9 | S8 1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The rules	15 min
Two negatives	25 min
Ordering	25 min
The symbols	20 min
Homework	5 min

LEARNING OBJECTIVES

- Compare any two integers using their positions on the line.
- Compare two negatives by comparing their moduli, in reverse.
- Order a list of integers, ascending and descending.
- Use the symbols <, >, ≤, ≥ correctly.

TEXTBOOK

National	pp. 59–63
Cambridge	Stage 8, pp. 2–10
Workbook	Exercise 1.1

01 Explanation

The rules

CASE	RULE	EXAMPLE
both positive	as usual	$3 < 8$
one of each	the negative is smaller	$-3 < 8$
a negative and zero	the negative is smaller	$-3 < 0$
zero and a positive	zero is smaller	$0 < 8$
both negative	the one with the larger modulus is smaller	$-8 < -3$

Every row is the same statement about the line: further right means greater.

ONE PICTURE REPLACES FIVE RULES

Sketch the line, mark both numbers, read left to right. The table above is only a record of what the picture shows.

Two negatives

This is the only case that catches people out.

> For negatives: the bigger the modulus, the smaller the number

PAIR	MODULI	COMPARISON
$-3, -8$	$3, 8$	$-8 < -3$
$-15, -2$	$15, 2$	$-15 < -2$
$-1, -100$	$1, 100$	$-100 < -1$

A debt of 100 is a worse position than a debt of 1; a temperature of $-100°$ is colder than $-1°$. The rule matches every real situation.

> $-8 > -3$ IS FALSE
>
> Comparing the moduli and forgetting to reverse is the standard mistake. Saying the numbers aloud as temperatures settles it in a second.

Ordering

Ascending means smallest first; **descending** means largest first.

LIST	ASCENDING
$3, -7, 0, -1, 5$	$-7, -1, 0, 3, 5$
$-2, -9, -5$	$-9, -5, -2$
$10, -10, 1, -1$	$-10, -1, 1, 10$

A method that never fails: put the negatives first, ordered by decreasing modulus, then zero, then the positives ordered as usual.

> SAY THE LIST AS TEMPERATURES
>
> “Minus nine, minus five, minus two, zero, three, five” is plainly a list of increasing warmth. The language does the ordering for you.

The symbols

SYMBOL	READ	TRUE EXAMPLE
$<$	is less than	$-5 < -2$
$>$	is greater than	$3 > -7$
$\leq$	is less than or equal to	$4 \leq 4$
$\geq$	is greater than or equal to	$-2 \geq -2$
$\neq$	is not equal to	$3 \neq -3$

A double inequality such as $-3 < x \leq 2$ says two things at once, and is read as a stretch of the line between two marks.

THE POINT OF THE SYMBOL FACES THE SMALLER NUMBER

$-5 < -2$: the narrow end is at -5. Reading the symbol as a mouth that eats the larger number is a reliable check.

02 Worked examples

EXAMPLE 1 **Compare -8 and -3, and -8 and 3.**

1. Both negative: compare moduli 8 and 3.
2. Larger modulus, smaller number: $-8 < -3$.
3. A negative is always less than a positive.
4. $-8 < 3$

ANSWER

$-8 < -3$ and $-8 < 3$

EXAMPLE 2 **Put $3, -7, 0, -1, 5$ in ascending order.**

1. Negatives first, by decreasing modulus: $-7, -1$.
2. Then zero.
3. Then the positives: $3, 5$.
4. $-7, -1, 0, 3, 5$

ANSWER

$-7, -1, 0, 3, 5$

EXAMPLE 3 **List all integers x with $-4 \le x < 2$.**

1. $\le$ includes -4.
2. $<$ excludes 2.
3. Count from -4 up to 1.
4. $-4, -3, -2, -1, 0, 1$

   Six integers.

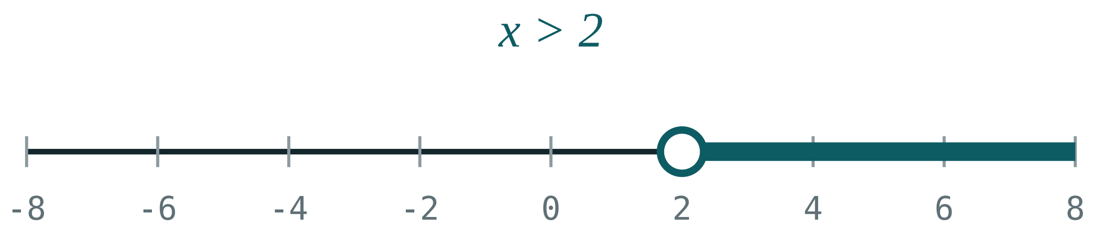
ANSWER

$-4, -3, -2, -1, 0, 1$

03 Interactive model

Read a week of winter temperatures aloud and ask which day was coldest; the ordering of negatives is settled by experience before it is a rule.

Ordering on the line



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
To compare	Taqqoslash	Сравнить
Greater than	Katta	Больше
Less than	Kichik	Меньше
Ascending order	O‘sish tartibi	По возрастанию
Descending order	Kamayish tartibi	По убыванию
Inequality sign	Tengsizlik belgisi	Знак неравенства
Between	Orasida	Между
Consecutive	Ketma-ket	Последовательные

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which is greater, -8 or -3?

- -8
- -3
- equal
- cannot tell

A negative number compared with a positive is:

- greater
- less
- equal
- it depends

For negatives, a larger modulus means:

- a larger number
- a smaller number
- the same
- nothing

Ascending order of $-2, -9, -5$:

- $-2, -5, -9$
- $-9, -5, -2$
- $-5, -2, -9$
- $-9, -2, -5$

$-2 \geq -2$ is:

- true
- false
- meaningless
- sometimes

Integers with $-4 \leq x < 2$ number:

- 5
- 6
- 7
- 4

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Greater: -8 or -3

   ANSWER -3

2. Greater: -8 or 3

   ANSWER 3

3. Greater: 0 or -1

   ANSWER 0

4. Greater: -100 or -1

   ANSWER -1

5. Is $-5 < -2$?

   ANSWER True

6. Is $-5 > -2$?

   ANSWER False

7. Is $4 \leq 4$?

   ANSWER True

Medium · the standard question

7 PROBLEMS

1. Ascending: $3, -7, 0, -1, 5$

ANSWER $-7, -1, 0, 3, 5$

2. Ascending: $-2, -9, -5$

ANSWER $-9, -5, -2$

3. Descending: $10, -10, 1, -1$

ANSWER $10, 1, -1, -10$

4. Integers with $-4 \le x < 2$

ANSWER $-4, -3, -2, -1, 0, 1$

5. Integers with $-2 < x \le 3$

ANSWER $-1, 0, 1, 2, 3$

6. The greatest integer less than -3

ANSWER -4

7. The least integer greater than -3

ANSWER -2

Hard · stretch and reason

7 PROBLEMS

1. How many integers satisfy $-20 < x < 15$?

   ANSWER 34

2. The greatest negative integer

   ANSWER -1

3. The least positive integer

   ANSWER 1

4. Order $-|-4|, |-3|, -(-2), 0$

   ANSWER $-4, 0, 2, 3$

5. If $a < 0$ and $b > 0$, compare a and $-b$

   ANSWER Cannot tell without more

6. Between which two consecutive integers does -3.7 lie?

   ANSWER -4 and -3

7. The smallest three-digit negative integer

   ANSWER -999

07 Homework

Homework — 5 tasks

Sketch the line for every comparison, and say each list aloud as temperatures.

1. Compare -11 and -4, and -11 and 4.
2. Put in ascending order: $-6, 2, 0, -13, 7$.
3. Put in descending order: $-1, -8, 5, 0$.
4. List all integers x with $-5 < x \leq 1$.
5. Name the greatest negative integer and the least positive integer.